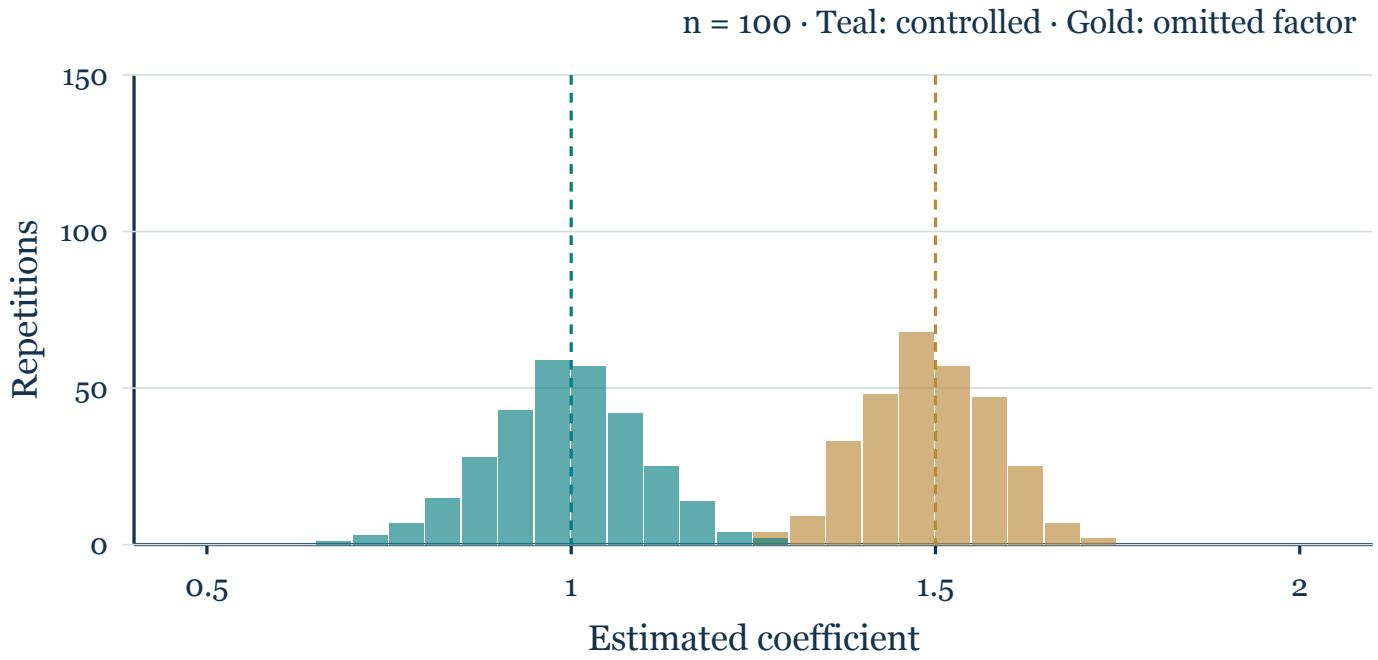
# Can a larger dataset rescue a biased design?

Use a known population to separate bias from sampling noise.




Actual Stata Monte Carlo estimates: 300 repetitions at each sample size. The data-generating staffing coefficient is 1; the omitted-variable target is 1.5.

# Define the simulation as an economic thought experiment

The generated truth is known by construction.

> $X = A + V$

> $Y = 1 + X + A + U$

> $A, V, U$ independent, with mean zero and variance one

# Unbiasedness and consistency ask different questions

One concerns an average at a sample size; the other concerns a limit.

> Unbiasedness: correct repeated-sample mean at a fixed sample size.

> Estimator consistency: concentration near the target as sample size grows.

> Causal consistency: observed/potential-outcome link; a different use of the word.

# The law of large numbers stabilises averages

More data can reveal a stable but unwanted relationship.

> Law of large numbers: sample averages approach population means under stated conditions.

> Consistent estimator: concentrates near its target as the sample grows.

# Measure co-movement before calculating the limit

Covariance records whether centered variables tend to move together.

> $\operatorname{Cov}(X, A) = E[(X - EX)(A - EA)]$

> $X = A + V: \quad \operatorname{Var}(X) = 2, \quad \operatorname{Cov}(X, A) = 1$

# Calculate the wrong limit before simulating it

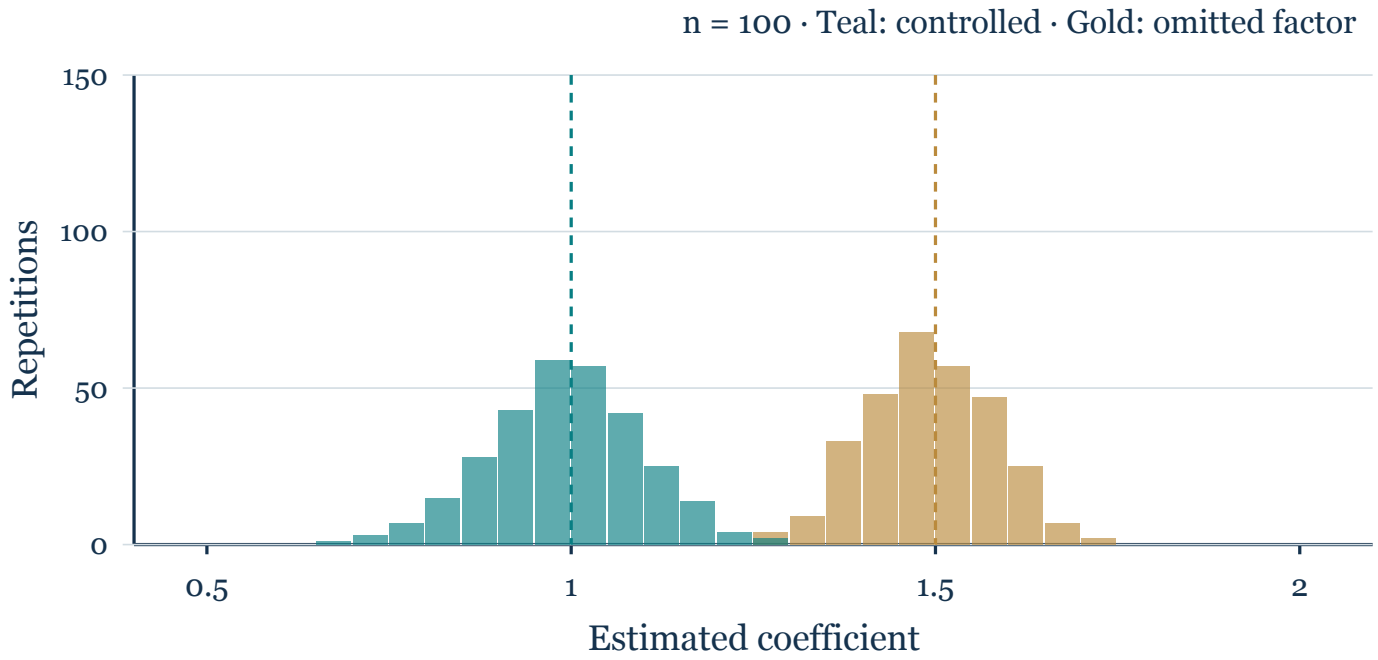
The short slope tends to one and a half.

$$\delta_1 = \frac{\operatorname{Cov}(X, A)}{\operatorname{Var}(X)} = \frac{1}{2}$$

$$\operatorname{plim} \widehat{b}_{\text{short}} = 1 + \frac{1}{2} = 1.5$$

# Compare repeated samples at two sizes

Concentration alone is not evidence of a correct target.




Actual Stata Monte Carlo estimates: 300 repetitions at each sample size. The data-generating staffing coefficient is 1; the omitted-variable target is 1.5.

# More data, same bias

Try the comparison before opening the explanation.

**Practice**

An estimator converges to 1.4 while the causal target is 1.0. What happens to the bias problem as n grows?

# More data, same bias: explanation

Follow the calculation and interpret its units.

> It remains: the estimator is inconsistent for the causal target. Sampling variation may shrink around the wrong limit. Narrow intervals around 1.4 do not repair the design.

# Read one large generated sample

A small SE can accompany substantial omitted-variable bias.

```
. regress y x, vce(robust)

------------------------------------------------------------------------------
             |               Robust
           y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           x |   1.516317   .0139733   108.51   0.000     1.488921    1.543712
       _cons |   .9947365   .0194616    51.11   0.000     .9565808    1.032892
------------------------------------------------------------------------------
```

# The central limit theorem describes scaled fluctuations

The square-root rate concerns sampling variation around a limit.

> Average of independent zero-or-one draws, each with mean 1/2: $SD(\overline{Y}) = 0.5/\sqrt{n}$

> Coefficient analogue: $\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, \Omega)$

# Measurement error creates a different source of inconsistency

A noisy regressor is not just a noisier outcome.

$$W = X + M, \quad Y = a + \beta W + (u - \beta M)$$

$$\operatorname{plim} \hat{b}_W = \beta \frac{\operatorname{Var}(X)}{\operatorname{Var}(X) + \operatorname{Var}(M)}$$

# Check attenuation in a separate generated comparison

Use a clean outcome equation and known noise variance.

```
. regress clean_y noisy_x, vce(robust)

------------------------------------------------------------------------------
             |               Robust
     clean_y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     noisy_x |   .5177554   .0112879    45.87   0.000     .4956249    .5398859
       _cons |   1.045476   .0221565    47.19   0.000     1.002037    1.088915
------------------------------------------------------------------------------
```

# Measurement error

Try the comparison before opening the explanation.

**Practice**

Observed x equals true $X^*$ plus mean-zero noise independent of $X^*$ and the outcome error. In a simple regression, what happens to the slope toward a positive true effect?

# Measurement error: explanation

Follow the calculation and interpret its units.

> Under classical measurement-error assumptions, the probability limit is $\beta \frac{\mathrm{Var}(X^*)}{\mathrm{Var}(X^*) + \mathrm{Var}(noise)}$, which is between zero and $\beta$. More observations do not remove this attenuation. The result does not automatically extend to nonclassical error or every multiple-regression setting.

# Outcome noise has another role

Mean-zero independent outcome error need not attenuate the slope.

> If the outcome is measured with an additional error unrelated to the regressors, that error enters the regression disturbance.

> The direction of measurement matters.

# Missing observations can change the population represented

Complete-case analysis needs a selection argument.

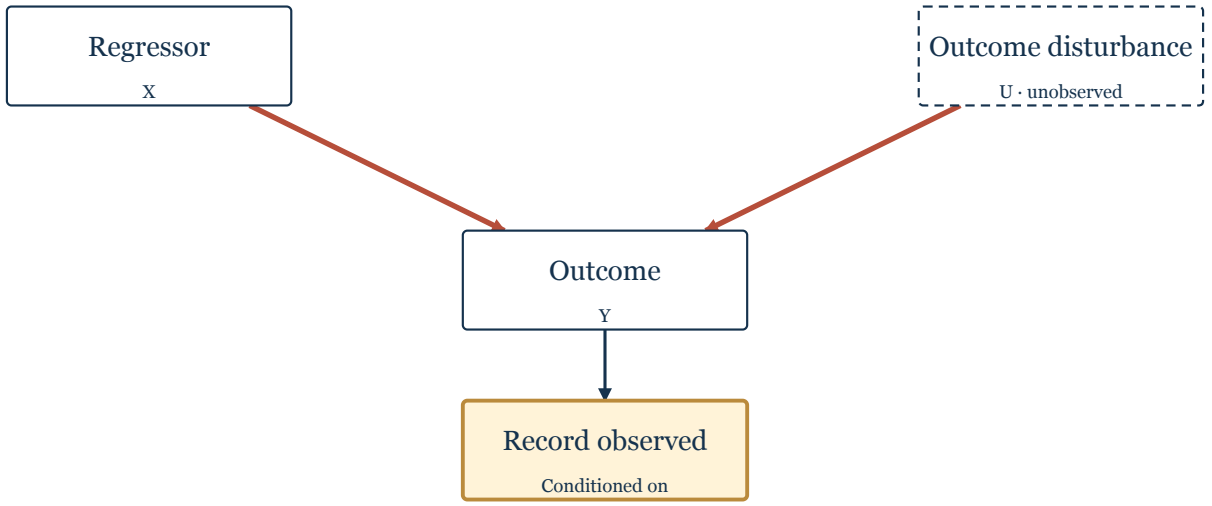



1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

# A missing-data assumption should be stated in context

Explain what makes the analysed cases informative.




0 open directed routes · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

# Leverage and influence are different

An unusual regressor value need not be a bad observation.

> Leverage: unusual regressor values.

> Large residual: unusual outcome given the fitted predictors.

> Influence: material change in the fitted result when the case changes.

# Influence

Try the comparison before opening the explanation.

**Practice**

A high-income district sits far from the remaining x values. Must it be deleted?

# Influence: explanation

Follow the calculation and interpret its units.

> No. A distant x value can have high leverage; influence also depends on its residual and its effect on the fit. Verify the observation, explain substantive relevance and show defensible sensitivity analysis. Delete a confirmed error for a documented reason, not because it changes significance.

# Sensitivity analysis answers a stated question

Change one defensible feature and explain the comparison.

> Sensitivity analysis examines how conclusions change under plausible alternative choices, such as a sample restriction, transformation or influential observation.

> For example, compare a fitted income relationship with and without one verified extreme-income record and show the range supported by each sample.

# The data audit belongs before the policy sentence

Units, ranges and recording rules affect interpretation.

> Check how variables were measured, whether their ranges are possible and whether units changed over time or across sources.

> Record sample restrictions and verify identifiers before merging data.

# Choose a remedy for the named problem

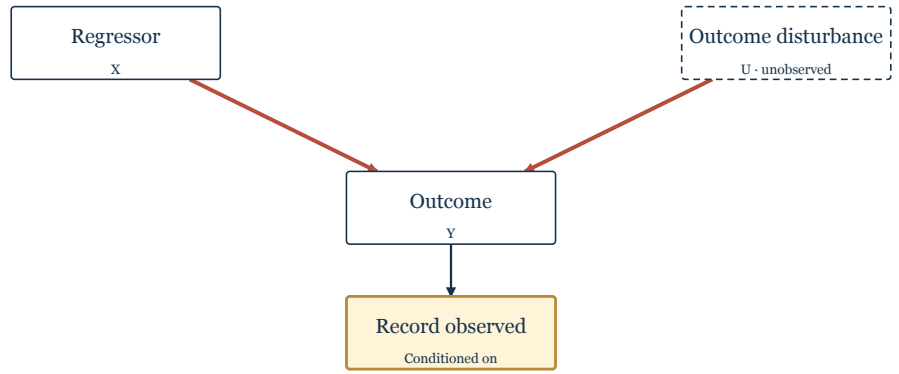
More observations, better measurement and better design do different jobs.

> If uncertainty is large under a credible comparison, more informative observations may help.

> State the problem before proposing a remedy.

# DAG practice: selection on an outcome descendant

Trace the routes before choosing a control.




0 open directed routes · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

## Practice

Suppose X and the outcome disturbance U are independent before selection, both affect Y, and a record is observed depending on Y. Why can restricting attention to observed records create an X–U association? Compare this with selection depending only on included X.

# DAG practice: selection on an outcome descendant: explanation

Explain what conditioning changes in this stated graph.

> Y is a collider on $X \to Y \leftarrow U$. The observation indicator S is its descendant. Conditioning on $S = 1$ can open the collider path, so the selected sample can violate the disturbance mean assumption even when the original sample satisfied it. If selection instead depends only on included X and carries no extra information about U conditional on X, this particular path is not opened. The conditional slope may remain valid under a correct model and adequate support, although precision and the represented population can change. The selection mechanism, rather than missingness alone, determines the concern.

# Separate simulation truth from estimated evidence

Reproduce the evidence, then explain its economic meaning.

**Practice**

Run meeting_06.do. Identify the known true coefficient in the first experiment, the short-regression probability limit, and the two estimated coefficients from the 4,000-observation data. Then explain the separate measurement-error result. Use the repeated-sample file to check whether a larger sample narrows the distribution around the correct or incorrect target.

# Separate simulation truth from estimated evidence: explanation

A reproducible result needs both a calculation and an interpretation.

> The first experiment sets the true direct X coefficient to one. Omitting A produces a probability limit of 1.5. In the generated 4,000-observation dataset, the short estimate is 1.516317 and the controlled estimate is 0.997692. They are estimates, so neither must equal its theoretical target exactly. The measurement-error experiment is generated separately. Its noisy-regressor estimate is 0.517755, compared with an attenuation limit of one half under its stated variances and independence assumptions. In the repeated-sample graphs, larger samples concentrate the short regression near 1.5 and the controlled regression near one. Precision improves around each model's limit; sample size does not change a biased design into a valid one.

# Next, compare groups without misreading interactions

A group difference changes which coefficient combination matters.

> The simulation separated stable sampling behaviour from a valid target, while the data examples showed why measurement and selection need their own arguments.

> The next lesson introduces indicators and interactions.